

Instagram Agent Hub

Setup guide and how to run the agents

YOU

approve every post / delete / reply

MAIN AGENT

plans the work, gives jobs to the 4 agents, merges the results.
The only one allowed to post or delete - and only after you type **yes**.

1. Reel Analyzer reel -> speech, hook, captions -> Word file	2. Content Writer script, captions, hashtags, how the reel should look	3. Viral Strategist plan for likes, comments, shares, followers	4. Performance Monitor first 3 hours: viral or not, then act
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1. What you need

Item	Needed for	Cost
A computer (Windows, Mac or Linux)	Running the agents	-
Python 3.10 or newer	Running the code	Free
ffmpeg	Reading video and audio from reels	Free
Git	Downloading the code from GitHub	Free
Claude API key (console.anthropic.com)	The brains of all agents	Pay per use
Instagram Business or Creator account + Meta developer app (optional)	Posting, deleting, live 3-hour monitoring, comment replies	Free

Tip: You can start **without** connecting Instagram. Reel analysis (Word file), content writing and the viral plan all work with just the Claude API key. The monitor then asks you to type the numbers from Insights.

2. Step-by-step setup

Step 1 - Install Python, ffmpeg and Git

System	What to do
Windows	Install Python from python.org/downloads (tick " Add python.exe to PATH " on the first screen). Install Git from git-scm.com . For ffmpeg, open PowerShell and run: winget install ffmpeg
Mac	Install Homebrew from brew.sh , then in Terminal: brew install python ffmpeg git
Linux	sudo apt install python3 python3-venv python3-pip ffmpeg git

Check everything is installed (close and reopen the terminal first):

```
python --version      # should say 3.10 or higher (on Mac/Linux use: python3 --version)
ffmpeg -version
git --version
```

Step 2 - Download the code

```
git clone https://github.com/pallavundhad/pallavundhad.git
cd pallavundhad
git checkout claude/instagram-agent-hub
```

Tip: Once the branch is merged into **main**, you can skip the git checkout line.

Step 3 - Create a virtual environment and install packages

A virtual environment keeps the agent's packages separate from the rest of your computer.

Mac / Linux

```
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
```

Windows (PowerShell)

```
python -m venv .venv
.\.venv\Scripts\Activate.ps1
pip install -r requirements.txt
```

You will see (.venv) at the start of the terminal line. **Every time** you open a new terminal to use the agents, go to the pallavundhad folder and run the activate line again.

Important: If PowerShell says scripts are disabled, run once: Set-ExecutionPolicy -Scope CurrentUser RemoteSigned

Step 4 - Add your Claude API key

- Go to **console.anthropic.com**, sign in, add a payment method under **Billing**.
- Open **API Keys** -> **Create Key**. Copy the key (starts with sk-ant-).
- Set it in your terminal:

Mac / Linux (this terminal only)

```
export ANTHROPIC_API_KEY="sk-ant-...your key..."
```

Windows PowerShell (this terminal only)

```
$env:ANTHROPIC_API_KEY = "sk-ant-...your key..."
```

To make it permanent: on Mac/Linux add the export line to ~/.zshrc or ~/.bashrc; on Windows run setx ANTHROPIC_API_KEY "sk-ant-..." and open a new terminal.

Important: Never share your API key or put it in a GitHub file. Anyone with it can spend your credit.

Step 5 (optional) - Connect your Instagram account

Needed only for **publishing, deleting, live monitoring and replying to comments** from the agents.

- **Switch to a professional account:** Instagram app -> Settings -> Account type and tools -> Switch to **Business** or **Creator**.
- **Link a Facebook Page** to the Instagram account (Instagram -> Edit profile -> Page, or from the Facebook Page settings).
- **Create a Meta app:** go to **developers.facebook.com** -> My Apps -> Create App, and add the **Instagram** product (Instagram API with Facebook Login).
- **Get a token:** open **Graph API Explorer** (Tools menu), choose your app, click **Generate Access Token** and grant: `instagram_basic`, `instagram_content_publish`, `instagram_manage_comments`, `instagram_manage_insights`, `pages_show_list`, `pages_read_engagement`.

- **Make it long-lived (about 60 days):** paste the token into the **Access Token Debugger** and click **Extend Access Token**.
- **Find your Instagram account ID:** in Graph API Explorer run `me/accounts?fields=instagram_business_account`. The number under `instagram_business_account -> id` is your **IG_USER_ID**.

Mac / Linux

```
export IG_USER_ID="17841400000000000"
export IG_ACCESS_TOKEN="EAAG...long token..."
```

Windows PowerShell

```
$env:IG_USER_ID = "17841400000000000"
$env:IG_ACCESS_TOKEN = "EAAG...long token..."
```

Tip: Meta changes its developer screens from time to time. If a button has moved, search the Meta docs for "Instagram Graph API get started". The token expires after about 60 days - repeat the Extend step then.

Step 6 (optional) - Let Agent 1 open reel links

Instagram often blocks downloads unless you are logged in. Two options:

- **Easiest:** save the reel video to your computer and give the agent the file path (e.g. `~/Downloads/reel.mp4`).
- **Links:** install a browser extension such as "Get cookies.txt LOCALLY", export your `instagram.com` cookies to `cookies.txt`, then set `IG_COOKIES_FILE=cookies.txt` (same way as the API key). Keep this file private - it is your login.

The first reel you analyse downloads the speech-to-text model (a few hundred MB). This happens once.

3. How to run the agents

Open a terminal, go to the project folder and activate the environment (Step 3). Then use one of these:

Command	What it does
<code>python -m instagram_agent_hub</code>	Chat with the main agent . It decides which agents to use.
<code>python -m instagram_agent_hub "your request"</code>	One request to the main agent, then continues as a chat.
<code>python -m instagram_agent_hub analyze <link or file></code>	Run Agent 1 only -> Word file.
<code>python -m instagram_agent_hub monitor <post ID></code>	Run Agent 4 only: the 3-hour check.

Tip: On Mac/Linux, if python is not found, use python3. Press **Ctrl+C** to quit the chat.

Main agent (chat) - example requests

```
you> Analyse https://www.instagram.com/reel/ABC123/ and write me a similar reel
      for my home bakery in Pune. I speak Hinglish, audience is women 22-35.

you> Make the script shorter, 20 seconds, and give 3 more hooks.

you> Plan how to get this reel viral. I have 4,000 followers.

you> Post it. Video: https://mycdn.com/bakery-reel.mp4  Caption: use option B.

you> Start the 3-hour monitor.
```

The main agent shows you its plan, runs the right agents (independent ones run at the same time), and gives you one combined answer. After every answer a full report is saved in reports/.

Agent 1 - Reel Analyzer (Word file)

```
python -m instagram_agent_hub analyze https://www.instagram.com/reel/ABC123/
python -m instagram_agent_hub analyze ~/Downloads/reel.mp4 --notes "focus on the hook"
```

The Word file (reports/reel-breakdown-...docx) contains:

Section	What's inside
1. Hook (first 3 seconds)	Spoken hook, on-screen text, visual, hook type, why it works
2. Speech	Tone, pace, energy, language style, delivery techniques, full timestamped transcript
3. On-screen captions	Every text overlay with its time
4. Posted caption	The original caption, how it is built, hashtags used
5. Structure	Beat-by-beat breakdown of the reel
6. Draft captions	3 new captions for you in different styles

Section	What's inside
7. Hook ideas	5 alternative hooks you can use
8. Key takeaways	What to copy into your own reels

Agent 2 - Content Writer and Agent 3 - Viral Strategist

These run through the main agent. Just ask in the chat, for example:

```
you> Write a 25-second reel script about 3 eggless cake mistakes, with how it
      should look, captions and hashtags.
you> How do I make this reel go viral? What should I do in the first hour?
```

Content Writer gives: 3 hooks, a timestamped script (what to say, text on screen, the shot), how the reel should look (framing, angles, lighting, transitions, text style, music, length, cover), 2-3 captions and hashtags. **Viral Strategist** searches the web for current trends and gives: before-posting fixes, best posting time, a first-60-minutes checklist, triggers for comments/saves/shares/follows, and target numbers for 1 h and 3 h.

Agent 4 - Performance Monitor (first 3 hours)

```
python -m instagram_agent_hub monitor 17912345678901234
python -m instagram_agent_hub monitor 17912345678901234 --checkpoints 30,60,120,180
```

- **Post ID:** the main agent shows it after publishing. For a reel you posted yourself, find it with Graph API Explorer: <IG_USER_ID>/media - the newest item is first.
- Checks run at **30, 60, 120 and 180 minutes after posting**. Keep the terminal and computer awake until the end.
- Each check gives a verdict - **VIRAL, ON TRACK, SLOW or FLOPPING** - with the numbers, the actions it took, and a prioritised list for you (Story share with a sticker, pin a comment, collab, part 2, or at 3 hours: re-post with a new hook).
- Not connected to Instagram? It asks you to type views, reach, likes, comments, shares and saves at each check.

Approving actions on your account

Whenever an agent wants to **post, delete, reply to a comment or hide a comment**, you will see:

```
=====
APPROVAL NEEDED: Publish a new reel
-----
Video: https://mycdn.com/bakery-reel.mp4

Caption:
3 mistakes that ruin eggless cakes ...
=====
Type "yes" to allow, anything else to cancel:
```

Safety: Only typing **yes** allows it. Anything else cancels and the agent continues without doing it. This check is built into the code, so no agent can skip it. The monitor never deletes or re-posts - it only recommends.

Important: Instagram only accepts a video for publishing from a **public link** to the .mp4 file (for example a file-hosting or cloud storage link that opens the video directly). If you don't have one, post from the app and then give the post ID to the monitor.

4. A typical day: from reference reel to 3-hour check

Step	You say / do	Who works
1	"Analyse this reel: <link>"	Agent 1 -> Word file
2	"Write my own version for <your niche>"	Agent 2 (+ uses the Word analysis)
3	"How do I make it viral?"	Agent 3 (web search)
4	Shoot and edit the reel using the script and look guidance	You
5	"Post it with caption B" -> type yes	Main agent (after your approval)
6	"Start the 3-hour monitor"	Agent 4 at 30 / 60 / 120 / 180 min
7	Do the actions it lists; approve or skip replies	You + Agent 4
8	Read the final summary and plan the next reel	Main agent

5. Where your files are saved and settings

File	Where
Word breakdowns from Agent 1	reports/reel-breakdown-<creator>-<date>.docx
Frames, audio and video used for analysis	reports/reel-<date>/
Full report of every chat answer	reports/report-<date>.md
3-hour monitor log	reports/monitor-<post ID>-<date>.md

Optional settings (set them like the API key):

Setting	Default	What it changes
HUB_MODEL	claude-opus-5-5	Claude model used by all agents
HUB_SUBAGENT_EFFORT	medium	How hard the 4 agents think: low (cheaper) / medium / high
HUB_ORCHESTRATOR_EFFORT	high	How hard the main agent thinks
WHISPER_MODEL	small	Speech-to-text accuracy: tiny / base / small / medium / large-v3. Use medium or large-v3 for Hindi / Hinglish (slower)
HUB_REPORTS_DIR	reports	Folder for all output files
IG_COOKIES_FILE	(none)	Cookies file so Agent 1 can open reel links
IG_GRAPH_VERSION	v23.0	Instagram Graph API version

6. Troubleshooting

Problem	Fix
python / ffmpeg not found	Reinstall it (Step 1) and open a new terminal. On Windows make sure "Add to PATH" was ticked.
No module named ...	Activate the environment (Step 3) and run <code>pip install -r requirements.txt</code> .
Authentication error / invalid API key	Check ANTHROPIC_API_KEY is set in this terminal and has billing enabled.
Agent 1 cannot download the reel link	Use cookies (Step 6) or save the video and pass the file path.
Transcript is empty	The reel may have only music. For Hindi/Hinglish try WHISPER_MODEL=medium.
"Instagram account not connected"	Set IG_USER_ID and IG_ACCESS_TOKEN (Step 5). Analysis and writing still work without them.
Instagram API error: token expired / invalid	Make a new long-lived token (Step 5) - they last about 60 days.
Instagram API error: permission	Re-generate the token and tick all permissions listed in Step 5.
Publishing fails	The video link must be public and point straight at the .mp4. Reels must meet Instagram's length/format rules.
Delete fails	Some accounts can't delete through the API - delete it in the Instagram app.
Monitor stopped early	The computer slept or the terminal closed. Run the monitor command again - checks whose time has passed run straight away, then it waits for the rest.
"no terminal to ask, so it was NOT done"	Approvals need you at the keyboard. Run the command in a normal terminal window.

7. Safety, cost and good practice

- **You are always in control:** nothing is posted, deleted, replied to or hidden without your typed **yes**.
- **Only white-hat growth:** the agents never suggest buying followers or likes, engagement pods or follow/unfollow, which can get an account restricted.
- **Other creators' reels:** use Agent 1 to learn from them - do not re-upload their videos or copy captions word for word.
- **Cost:** you pay Anthropic per use. A full run (analysis + script + viral plan) uses several agent calls; the 3-hour monitor makes 4 more. Set HUB_SUBAGENT_EFFORT=low to spend less, and check usage at console.anthropic.com.
- **Keep secrets private:** API key, Instagram token and cookies.txt must never be shared or uploaded to GitHub.

More details: `instagram_agent_hub/README.md` in the project folder.